MIL-D-45413B
30 April 1975
SUPERSEDING
MIL-D-45413A
15 August 1962

MILITARY SPECIFICATION

DYNAMITE, MILITARY, M1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a medium velocity dynamite designated as M1 Military Dynamite.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Film (Polyethylene Film, Thin Gage)
PPP-B-26	- Bag, Plastic, Polyethylene, General Purpose
PPP-T-76	- Tape, Pressure-Sensitive Adhesive, Water Resistant, for Carton Sealing

MILITARY

MIL-T-248	- Trinitrotoluene (TNT).
MIL-B-2427	- Boxes, Ammunition Packing, Wood, Nailed.
MIL-L-10547	- Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-R-13742	- RDX, Desensitized.
MIL-P-13822	- Polyisobutylene/Oil Gel.
MIL-C-14003	- Cap, Blasting Special.
MIL-P-14796	- Process Oil.
MIL-C-17124	- Cord, Detonating.
MIL-G-51133	- Gum (Guar).
MIL-B-46505	- Box Ammunition Packing, Wood, Wirebound.
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For.
MIL-F-50449	- Filler Sheet Form, (For Use in Ammunition, Containers).

FSC: 1375

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-650 - Explosive: Sampling, Inspection and Testing.

DRAWINGS

8796522 - Marking Diagram and Sealing for Wooden Packing Boxes.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

The Interstate Commerce Commission Rules are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402. Orders for the above publication should cite "49 CFR Part 100-199, (latest revision)".

3. REQUIREMENTS

3.1 Materials. The materials used in the manufacture of military dynamite shall comply with the following specifications.

MIL-T-248 - Trinitrotoluene (TNT), type 1 crystalline.
MIL-R-13742 - RDX, desensitized.
MIL-P-13822 - Polyisobutylene/Oil Gel
MIL-P-14796 - Process Oil.
MIL-G-51133 - Gum (Guar).

The suggested composition of the Military Dynamite shall be a uniform mixture of the following materials:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

	Percent
Desensitized RDX	78
Trinitrotoluene (TNT)	15
Guar Gum	5
Polyisobutylene/Oil Gel	2

3.2 Cartridge. The explosive composition shall be packed in cylindrical paper shells forming cartridges with a nominal diameter of 1-1/4 inches and a nominal length of 8 inches. The cartridge shall be made from plain manila paper weighing approximately 70 pounds per ream (500 24-inch by 36-inch sheets) and shall be sprayed or dipped in paraffin before being filled. The cartridges shall be marked in black letters as follows:

DYNAMITE, MILITARY, M1
(Medium Velocity)
DANGER

3.3 Composition. The explosive composition shall be as follows when determined as specified in 4.5.1.

RDX	75 ± 1.0 percent
Trinitrotoluene (TNT)	15 ± 0.5 percent
Process oil plus Polyisobutylene	5 ± 0.5 percent
Guar Gum	5 ± 0.5 percent

3.4 Consistency. The explosive composition shall be sufficiently plastic at -70 degrees Fahrenheit (°F) to allow a cohesive mass to be formed by hand pressure when tested as specified 4.5.2.

3.5 Sensitivity.

3.5.1 Blasting cap and detonating cord. A column of military dynamite cartridges shall detonate throughout its entire length when tested as specified in 4.5.3.1 and 4.5.3.2.

3.5.2 Rate of detonation. The rate of detonation shall be not less than 17,716 feet per second and the rate of detonation, measured in the last cartridge in the column, shall not differ by more than 10 percent from the rate of detonation in the first cartridge in the column when determined as specified in 4.5.3.1 and 4.5.3.2.

3.5.3 Sensitivity to impact. The explosive composition shall not detonate when struck by a bullet from an M2 .30 caliber cartridge when tested as specified in 4.5.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

3.6 Water immersion. The military dynamite shall function high order after immersion in water for 24 hours when tested as specified in 4.5.5.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection (see 4.3).
b. Quality Conformance Inspection (see 4.4).
c. Packing Inspection (see 4.4.2 and 5).

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of 100 cartridges of military dynamite obtained by sampling as described in 4.4.2. The samples shall be obtained from a production batch which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.4.2 and 4.5 of this specification.

4.3.3 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of military dynamite shall contain explosive composition (see 3.1) from one production lot only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

4.4.2 Examination. Unless otherwise specified in the Classification of Defects and test tables, sampling plans for the major and minor defects shall be in accordance with MIL-Std-105, Inspection Level II (see MIL-A-48078 paragraph 4.4.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cartridge (see 3.2)	1 OF 1	NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101	Cylindrical paper shells		0.40	3.2	Visual
102	Diameter of shell		0.40	3.2	Gage
103	Length of shell		0.40	3.2	Gage
104	Type of paper		0.40	3.2	Visual
105	Weigh of ream		0.40	3.2	Scale
106	Coated with paraffin wax		0.40	3.2	Visual
107	Marking		0.40	3.2	Visual
Minor:	None defined.				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1.1	Unit package (see 5.1.1) Prior to closing	1 OF 1	NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101	Bag improper (Not coated paper or 0.005 inch polyethylene)		0.25	5.1.1	Visual
102	Number of cartridges incorrect		0.25	5.1.1	Visual/Manual
Minor:	None defined.				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1.2	Unit package (see 5.1.1) Closing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101	Seams open or torn		0.40	5.1.1	Visual
102	Metal clip or sealing strip improperly applied		0.40	5.1.1	Visual
103	Bag cut, punctured or torn		0.40	5.1.1	Visual
Minor:	None defined.				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Box, packing (see 5.2.1)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major:					
101	Box improperly assembled		0.40	5.2.1	Visual/Manual
102	Board broken or split		0.40	5.2.1	Visual/Manual
103	Strapping (nailed wood boxes) or wires (wirebound boxes) missing, broken or loose		0.40	5.2.1	Visual/Manual
104	Box damaged so that contents are exposed or liable to become exposed		0.40	5.2.1	Visual
Minor:					
201	Strapping improperly assembled (nailed wood bixes) or wire loops not properly fastened (wirebound boxes)		1.50	5.2.1	Visual/Manual
202	Contents loose when box is tilted		1.50	5.2.1	Manual
203	Marking misleading or unidentifiable (see 5.2)		1.50	5.2.1	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

4.4.3 Testing

4.4.3.1 Sampling

4.4.3.1.1 Sampling for cartridge. - Unless otherwise specified, two (2) unit packages containing fifty (50) cartridges shall be selected at random from each lot. The sample cartridges shall be subjected to the tests specified in 4.5.3 thru 4.5.5. If any sample fails to comply with the specified requirement given in Section 3, the lot shall be rejected.

4.4.3.1.2 Sampling for composition. - Remove the composition from two dynamite cartridges selected at random to form two independent samples. Perform the tests specified in 4.5.1 on each of these samples. If either sample fails to comply with the specified requirement given in Section 3, the lot shall be rejected.

4.4.3.1.3 Sampling for consistency. - Remove the composition from three (3) dynamite cartridges selected at random to form three independent samples. Perform the test specified in 4.5.2 on each of these samples. If any sample fails to comply with the specified requirement given in Section 3, the lot shall be rejected.

4.4.3.2 Testing of military dynamite cartridges and composition. - The samples obtained in 4.4.3.1.1 thru 4.4.3.1.3 shall be subjected to the following tests.

Inspection Requirement	Defect Classification
Composition	
RDX (see 3.3)	Major B
TNT (see 3.3)	Major B
Process Oil (see 3.3)	Major B
Guar Gum (see 3.3)	Major B
Consistency (see 3.4)	Major B
Sensitivity (see 3.5)	Major B
Blasting Cap (see 3.5.1)	Major B
Rate of Detonation (see 3.5.2)	Major B
Sensitivity to Impact (see 3.5.3)	Major B
Water Immersion (see 3.6)	Major B

4.4.4 Inspection equipment. The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

inspection equipment shall be employed, where applicable for all tests and examinations specified in 4.5. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes.

4.5 Test methods and procedures. (see 6.3) The tests in 4.5.1 shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical textbooks.

4.5.1 Determination of composition. A portion of approximately 2.0 grams of the sample of the explosive composition shall be accurately weighed and transferred to a 100 ml beaker. Prepare a mixture of Freon 113/Chloroform by mixing 200 ml Freon 113 (1,1,2-Trichlorotrifluoroethane) with 100 ml of chloroform. Thirty ml of Freon 113/chloroform shall be added and covered with a watch glass. The beaker and contents shall be placed on a steam bath for one half four. The beaker and contents shall be removed from the steam bath and allowed to cool to room temperature. With the aid of a Fisher filtrator, the supernatant liquid contained in the beaker shall be filtered. The residue in the beaker shall be transferred quantitatively to a filtering crucible with a fine stream of Freon 113/chloroform from a wash bottle. The crucible and contents shall be washed with two 20 ml portions of Freon 113/chloroform. The crucible and contents shall be aspirated until the odor of Freon 113/chloroform is no longer discernible. The crucible and contents shall be dried in an oven maintained at 70 + 5°C for 1 hour, cooled in a desiccator and weighed. The weight shall be recorded and the crucible with the contents shall be reserved for the RDX and guar gum determinations. The filtrate shall be reserved for the TNT determination.

4.5.1.1 TNT determination. The filtrate obtained in the extraction specified in 4.5.1 shall be transferred quantitatively to a 250 ml volumetric flask. The beaker shall be washed carefully with the aid of a stream of Freon 113/chloroform, transferring all washings to the volumetric flask. The volumetric flask shall be filled to the mark with additional Freon 113/chloroform. A calibrated 25 ml pipette shall be rinsed with the Freon 113/chloroform solution of the sample and then a 25 ml portion of the solution shall be transferred to a titration flask. The Freon 113/chloroform shall be evaporated to dryness with a slow stream of dry air. Twenty-five ml of glacial acetic acid shall be added to the flask to redissolve the residue. The TNT shall be determined in accordance with method 413.1 of MIL-STD-650 beginning with paragraph 5.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

4.5.1.2 RDX. The crucible reserved from 4.5.1 shall be washed with 20 ml of acetone, and the acetone shall be allowed to drip through before applying suction. The solution shall be aspirated and the above procedure shall be repeated until a total of five 20 ml extractions have been made. The sides of the crucible shall be washed with additional 20 ml portions of acetone, until the following qualitative test for RDX in the acetone washings is negative: A small portion of acetone filtrate shall be transferred to a 25 ml beaker, evaporated to dryness on a steam bath, and several drops of a solution of chromotropic acid in 36 normal sulfuric acid shall be added. The formation of a reddish-brown color indicates the presence of RDX. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in an oven maintained at 70 ± 5°C for one hour, cooled in a desiccator, and weighed. The percentage of RDX shall be calculated to determine compliance with 3.3 as follows:

$$\text{Percentage of RDX} = \frac{(A-B)\ 100}{W}$$

where:

A = weight of crucible plus contents before acetone extraction, gram

B = weight of crucible plus contents after acetone extraction, grams

W = weight of sample, grams

4.5.1.3 Guar gum. The percentage of guar gum shall be calculated to determine compliance with 3.3 as follows:

$$\text{Percentage Guar Gum} = \frac{(A-B)\ 100}{W}$$

here:

A = weight of crucible after acetone extraction, grams

B = weight of empty crucible, grams

W = weight of sample, grams

4.5.1.4 Polyisobutylene and process oil. The percentage of polyisobutylene and process oil shall be calculated to determine compliance with 3.3 by adding the percentage of TNT, RDX and guar gum and subtracting the sum from 100 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

4.5.2 Consistency. Explosive composition removed from the sample cartridges, shall be transferred to individual beakers or other suitable containers, covered and placed in a cold box maintained at -70 + 5°F for 24 hours. The hand moldability to form a cohesive mass at this temperature shall be determined by working the material with the hands to determine compliance with 3.4.

4.5.3 Sensitivity.

4.5.3.1 Blasting cap. Sample cartridges shall be placed end to end in contact on smooth dry ground to form a column approximately 4 feet long. One end of the column of cartridges shall be initiated by a blasting cap complying with MIL-C-14003. Observation shall be made for detonation of the cartridges throughout the entire length of the column to determine compliance with 3.5.1. The rate of detonation shall be determined in the first and last cartridges of the column by the counter chronograph or dautriche method to determine compliance with 3.5.2.

4.5.3.2 Detonating cord. Sample cartridges shall be placed end to end in contact on smooth dry ground to form a column approximately 4 feet long. Detonating cord complying with MIL-C-17124, type I, class C, shall be attached to the dynamite cartridge one inch from the end by a girth hitch and one extra wrap through the girth hitch. One end of the column shall be initiated by the detonating cord. Observation shall be made for detonation of the cartridges, throughout the entire length of the column to determine compliance with 3.5.1. The rate of detonation shall be determined in the first and last cartridges of the column by the counter chronograph or dautriche method to determine compliance with 3.5.2. If the initiating device interferes with the measurement of the rate of detonation in the first stick, the following modification may be used in 4.5.3.1 and 4.5.3.2. Add an additional stick of dynamite to the column and determine the rate of detonation in the second and last cartridge of the column instead of the first and last cartridge.

4.5.4 Sensitivity to impact. Explosive composition, removed from sample cartridges, shall be packed to its normal density to form a 3 inch by 3 inch by 1 inch block. A 3/8 inch thick mild steel plate shall be placed against one of the 3 inch by 3 inch faces of the block. A bullet from an M2.30 caliber ball cartridge shall be fired at a range of approximately 100 feet into the other 3 inch by 3 inch face of the block. Observation shall be made to determine compliance with 3.5.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

4.5.5 Water immersion. Five sticks of dynamite shall be immersed in 12 feet of water for a period of 24 hours. After 24 hours, the sticks shall be removed from the water and initiated as specified in 4.5.3.1, except that the rate of detonation shall not be determined.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.4).

5.1.1 Level A

5.1.1.1 Fifty (50) cartridges shall be packaged in a polyethylene bag constructed in accordance with PPP-B-26, except that it shall be a plain flat or gussetted bag without wire-tie or interlocking seal. Bag material shall comply with Type II Class 1 of L-P-378, grade and finish optional, and film thickness shall be .005 inch. The bag shall be closed by gathering the top of the bag together to form a neck and closing the neck securely with a wire clip clinched into a ring with ends of the ring overlapping not less than 5/16 inch. The wire shall have a nominal diameter of 0.118 inch and shall be of aluminum alloy material. The clip shall be applied to the bag tight enough to prevent any slippage but shall not cut, puncture, or damage the bag material.

5.1.1.2 Alternate packaging. Fifty (50) cartridges shall be packaged in a waterproof bag conforming to MIL-L-10547 for a Type I, Grage B liner. Excess air shall be removed from the bag and the bag closed by bringing the top edges of the bag together and folding the bag walls down flat on the contents by a series of complete turns (drugstore fold). A length of 2 inch wide PPP-T-76 pressure-sensitive tape shall be applied over the full length of the fold and extended down the sides of the bag to retain the top and bottom bag ears firmly against the bag sides.

5.1.2 Level C. Cartridges shall be packaged in accordance with the requirements of the Code of Federal Regulations, CFR 49, Parts 100-199.

5.2 Packing.

5.2.1 Level A. Two unit packages (100 cartridges) shall be packed in a wooden box conforming to MIL-B-2427, Grade A, Type I, Class 3 requirements of MIL-B-46506 for a type I class 1 Grade A Box. The interior sides, ends, top and bottom of the box shall be lined with fillers

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

sheet form conforming to MIL-F-50449 requirements. Minimum dimensions of end, side and bottom fillers shall be equal to the inside box dimensions less 1/2 inch. Top fillers shall be equal to the inside length and width of the box less 1/8 inch. Additional fillers shall be added as required to eliminate looseness of box contents. Wire-bound boxes shall be assembled and closed in accordance with the specification. Boxes shall be closed and strapped in accordance with instructions contained in the appendix of the box specification except flat strapping shall be zinc coated No. 16 (.0625) gate for overseas use. Two straps shall be applied to each box.

5.2.2 Level C. Cartridges shall be packed in accordance with the Code of Federal Regulations 49 CFR parts 100-199, and in a manner that will insure safe delivery from the manufacturer to the first receiving activity.

5.3 Marking. Boxes shall be marked in accordance with Drawing 8796522. The descriptive nomenclature shall be "100 CARTRIDGE, DYNAMITE, M1". The DOT marking "HIGH EXPLOSIVES-DANGEROUS" shall appear on the box cover and side above the descriptive nomenclature. In addition, the words "THIS SIDE UP" shall appear on the top of the box.

6. NOTES

6.1 Ordering data. See MIL-A-48078

6.2 Intended use. Military dynamite is intended for use where a medium velocity blasting explosive is required. It is to be used in military construction, quarry applications, and service demolition.

6.3 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-P, Dover, New Jersey 07801. This description should include, but not be limited, to the accuracy and precision of the method, test data to demonstrate the accuracy and precision, and drawings of any special equipment required.

6.4 Repacking of military dynamite. The following drawings should be used for procurement of box, carton and related packing materials only when repacking military

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.

dynamite; drawings 9227394, 9227393, and 9227592. This packing should not be used in conjunction with new procurement.

Custodian:
Army-PA
Navy-OS
Air Force-70

Preparing Activity:
Army-PA

Review Interest:
Army-PA
Navy-OS
Air Force-70

User Interest
Navy-YD
Project Number: 1375-0117

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:56Z
Check the source to verify that this is the current version before use.